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Modeling and Simulation of Spatial Characteristics in Epidural Electrical Stimulation for Targeted Neural Activation

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***Abstract*— Neuromodulation via epidural neural interfaces holds promise as a treatment approach for various neurological conditions affecting the human central nervous system. Epidural electrical stimulation has been shown to be effective in suppressing chronic pain and restoring motor functions. However, the quantitative understanding of the efficacy and spatial selectivity of epidural electrical stimulation (EES) through epidural neural interfaces remains limited. In this study, we employed computational modeling and simulation techniques to investigate the spatial characteristics of epidural neuromodulation for targeted neural activation. We constructed an anatomically realistic model of the human spinal cord and analyzed the distribution of induced electric fields for different stimulation configurations. Axon models embedded in white matter and nerve roots were used to simulate and evaluate evoked neural responses. Simulation results indicate that the spatial selectivity of EES does not consistently increase with decreasing contact spacing. Our findings provide valuable insights into optimizing epidural neural interfaces and offer quantitative guidance for improving the precision and efficacy of therapeutic applications.**

***Clinical relevance*—This study advances the understanding of EES for neuromodulation, with direct implications for clinical applications in treating chronic pain and motor dysfunction. By providing a detailed computational model of EES, the work offers insights into optimizing the spatial selectivity of neural activation, which is crucial for targeting specific neural structures while minimizing unwanted side effects. These findings can guide the design of more effective and precise epidural neural interfaces, enhancing the efficacy of EES therapies for various neurological conditions and potentially improving patient outcomes.**

I. INTRODUCTION

Epidural electrical stimulation (EES) has rapidly advanced in recent years as a therapeutic approach to alleviate chronic pain, facilitate neurological recovery, and support rehabilitative training (Fig. 1). Widely utilized for suppressing neuropathic pain [1]–[3], EES also holds promise for patients with spinal cord injuries (SCI) [4]–[16], stroke [17], and other conditions [15], [18], aiming to restore motor function by delivering electrical pulses through a neural interface implanted in the epidural space of the spinal cord. Muscles innervated by the spinal nerve roots would response with different patterns due to the activation of different structures and regions of the spinal cord [11], [19]. The spatial selectivity of electrical stimulation directly affects the flexibility and precision of muscle responses, which further affects the patient's rehabilitation outcome. Poor stimulation selectivity can lead to unwanted activation and even side effects for the patient. EES employ neural interfaces implanted in the epidural space, separated from target neural tissue by cerebrospinal fluid (CSF). The presence of the dura mater and highly conductive CSF can lead to diffused electrical stimulation, impacting the spatial selectivity of stimulation to neural tissue [20]. Moreover, varying stimulation configurations significantly influence stimulation outcomes [21]–[24].

In this study, we conducted a computational investigation to explore the spatial selectivity and efficacy of epidural neuromodulation of the spinal cord through modeling and simulation. First, we developed a comprehensive model of the human spinal cord based on anatomical data [25]–[30], incorporating gray matter, white matter, CSF, dura mater, and nerve roots (Fig. 2a). Electrode arrays with varied configurations were modeled and positioned in the epidural space, targeting the lumbar enlargement of the spinal cord. Electric fields induced by EES were derived using the finite element method (FEM). To assess the spatial selectivity of stimulation, we defined a selectivity index based on two criteria: activating function (the second spatial derivative along the direction of the membrane) [31]–[33] and potential. We simulated neural activities evoked by EES by modeling nerve fibers in white matter and nerve roots of the spinal cord using an axon-cable model [34]–[36]. Selectivity indices were calculated for stimulation configurations with different polarity combinations and contact spacing. To elucidate the spatial properties of EES, we generated selectivity index maps and threshold maps by varying the position of active electrodes. Considering the diversity of stimulation targets, we adjusted the size of target regions and stimulated them using electrode arrays with different contact spacing. The simulation results suggest a potential limit to spatial selectivity for EES, indicating that increasing the contact density of the neural interface (contact spacing less than 1 mm) may not necessarily achieve more selective stimulation. This computational study provides quantitative insights into EES and offers valuable guidance for optimizing neural interfaces.

II. MATERIALS AND METHODS

A. Anatomically Realistic Modeling of the Human Spinal Cord

We constructed an anatomically realistic model of the human spinal cord based on published anatomical data [25]–[30]. The model includes gray matter, white matter, cerebrospinal fluid (CSF), dura mater, and nerve roots spanning from segments C1 to S5 (Fig. 2a). The lengths and diameters

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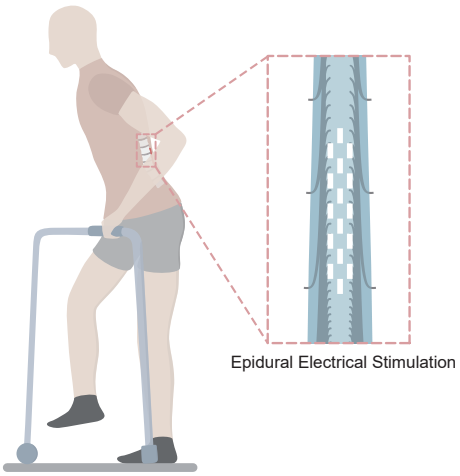


Fig. 1. Epidural electrical stimulation for motor function rehabilitation.

of the segments were adjusted according to the reported anatomical measurements [30], while the outlines of the gray matter were derived from an anatomical atlas [27]. The trajectories of the dorsal and ventral nerve roots were determined based on their entry points in the spinal cord relative to the corresponding intervertebral foramina [30]. Rootlet bundles at each nerve root's entrance were modeled by uniformly distributing rootlets across the segment, with the number of rootlets obtained from previous studies [37]. The dura mater diameter was estimated from vertebral canal diameters for different segments [38], with a thickness of 0.5 mm. The final model encompasses spinal segments C1 to S5 and the cauda equina, with a total length of 647.33 mm.

To simulate neural activity induced by epidural electrical stimulation (EES), nerve fibers in the white matter and nerve roots were modeled. We applied the Lloyd relaxation method [39] to uniformly distribute trajectories in the varying cross-sectional areas of the white matter [13], [20]. Trajectories were selected uniformly across axial cross-sections, with Lloyd relaxation used to redistribute randomly placed points, ensuring uniformity within the white matter region. Points located outside the white matter in the new sections were removed, and the remaining points were redistributed across all sections. Interpolation was used to generate continuous nerve fiber trajectories along the spinal cord. Neural responses were simulated using a realistic biophysical model, employing the McIntyre-Richardson-Grill (MRG) model to describe the active membrane dynamics at the nodes of Ranvier and passive internodal segments [36]. Axonal responses across different diameters were calculated to comprehensively assess the effects of stimulation.

B. Evaluation of Spatial Selectivity of EES

To assess the spatial selectivity of EES, we introduced a Selectivity Index (SI) to quantify the concentration of the electric field distribution:

$$\mathrm{SI} = \frac{V_{\mathrm{AT}}/V_{\mathrm{T}}}{1+(V_{\mathrm{A}}-V_{\mathrm{AT}})/V_{\mathrm{T}}} \tag{1}$$

Here, V_{T} represents the volume of the target region, V_{AT} denotes the activated volume within the target region, and V_{A} is the volume of all activated regions. We calculated the activated volume using two criteria: the activating function (AF-based SI) and the electric potential (Potential-based SI). The activating function, commonly used in Deep Brain Stimulation (DBS) simulations [32], [33], [40], [41], is defined as the second spatial derivative of extracellular voltage along an axon [31]. Positive values of the activating function are associated with a higher probability of activation induced by extracellular stimulation. Additionally, we used the electric potential as a criterion, defining regions where the electric potential falls below a threshold (commonly used for cathodic stimulation in EES) as activated [42]. The activated volume in the spinal cord is calculated by:

$$V_A = \iiint f(x,y,z)\mathrm{d}x\mathrm{d}y\mathrm{d}z \tag{2}$$

where $f(x,y,z)$ is the discriminant function that determines whether the position in the spinal cord is activated or not. Depending on the selected criterion, the discriminant function at the given location is defined as:

$$f(x,y,z) = \begin{cases} 1, & \text{if } V_e(x,y,z) \le V_{\text{threshold}} \\ 0, & \text{if } V_e(x,y,z) > V_{\text{threshold}} \end{cases} \tag{3}$$

or

$$f(x,y,z) = \begin{cases} 1, & \text{if } AF(x,y,z) \ge AF_{\text{threshold}} \\ 0, & \text{if } AF(x,y,z) < AF_{\text{threshold}} \end{cases} \tag{4}$$

$V_e(x,y,z)$ is the potential at the given position and $AF(x,y,z)$ is the second spatial derivative of the potential along the rostro-caudal direction. In concrete, we specified virtual $V_{\text{threshold}}$ and $AF_{\text{threshold}}$. Under the quasi-static assumption, the stimulation amplitudes were varied to obtain the highest SI.

The numerator of the definition of SI reflects the ratio of activated volume within the target region, whereas the denominator term accounts for the ratio of activated volume outside the target region relative to the total volume of the target. The defined selectivity index ranges from 0 to 1, where a value of 0 indicates complete activation within the target region with no activation outside of it, and a value of 1 represents no activation within the target region.

C. Simulating Electric Field Distribution and Neural Responses evoked by EES

We calculated the electric fields induced by EES using the finite element method (FEM). The 3D model of the human spinal cord was imported into COMSOL Multiphysics 5.5 for FEM calculations. Electrode arrays with varying contact spacing were automatically modeled in COMSOL by adjusting geometry parameters. Contacts were represented as square sheets positioned outside the dura mater. To replicate the *in vivo* environment, the spinal cord and electrode array

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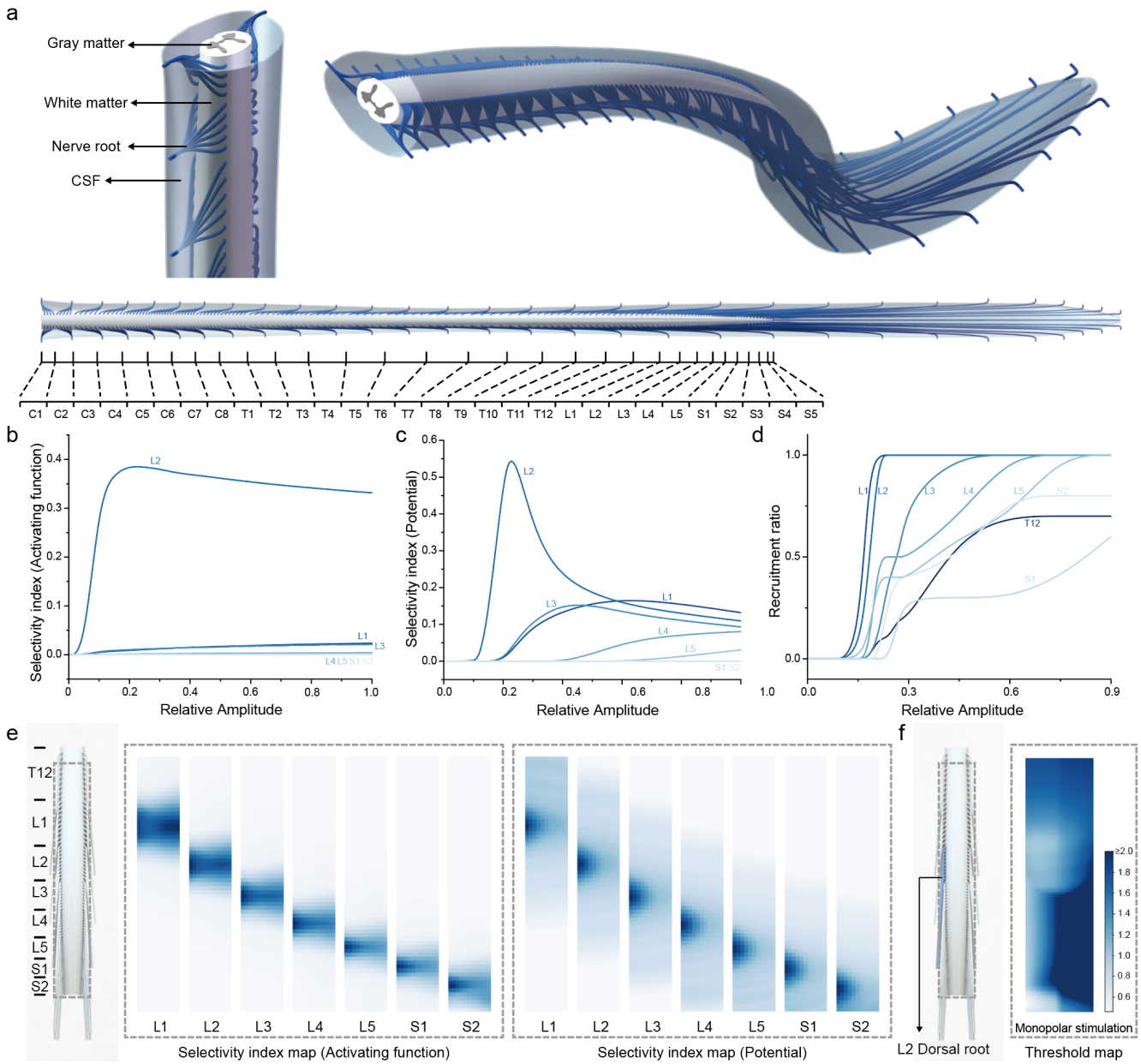


Fig. 2. Modeling and simulation of human EES. (a), Anatomically realistic model of the human spinal cord. (b), An example of AF-based SI curve with increasing stimulation amplitudes. The curve is generated with monopolar configuration targeting L2 segment. (c), Potential-based SI curve with increasing stimulation amplitudes. (d), The recruitment ratio of different nerve roots for the monopolar configuration targeting left L1 and L2 dorsal root. (e), Selectivity index maps for different segments of the spinal cord using monopolar configuration. Deeper colors indicate higher SI. (f), Threshold map of the left L2 dorsal root.

were placed in a saline cylinder [20], [36], [43], [44]. A nonuniform, second-order tetrahedral mesh was generated for model discretization, with tissue conductivity values obtained from literature sources [45] (Table I). A current source was applied at the cathode surface, with anodes grounded. Dirichlet boundary conditions were set at the outermost model boundaries to simulate grounding of the outer shell of the implantable pulse generator (IPG) at the distance:

$$V_e(\delta\Omega) = 0 \tag{5}$$

where $\delta\Omega$ represents the outermost surface (saline) of the model, and V_e is the potential. Electric potentials at each position were derived by solving the quasi-static approximation of the Maxwell Equations:

$$\Delta \cdot \sigma \Delta V_e = 0 \tag{6}$$

where σ is the conductivity tensor.

We focused on several clinically relevant stimulation configurations: monopolar stimulation (cathodic stimulation with the IPG shell grounded), multipolar stimulation (one cathode surrounded by six anodes), and bipolar stimulation with

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TABLE I
CONDUCTIVITY VALUES OF DIFFERENT TISSUES

Tissue	Conductivity (S/m)
White matter (longitudinal)	0.6
White matter (transversal)	0.083
Gray matter	0.23
CSF	1.65
Saline	2.0
Dural mater	0.04

different orientations (one cathode and one anode) (Fig. 3a). Contact spacing was varied between 1 mm and 3 mm to assess its impact on spatial selectivity. The cathode position in the lumbosacral epidural space, which is critical for lower extremity movements, was systematically traversed to generate selectivity index maps (Fig. 2e). Finite Element Method (FEM) solutions for each EES configuration were imported into MATLAB, where the selectivity index was calculated through integral operations. Stimulation amplitudes were adjusted for each configuration to maximize the selectivity index. Scalar factors were applied to the FEM solutions to simulate different stimulation amplitudes under quasi-static assumptions [20], [46].

For each EES configuration, extracellular voltages were applied to MRG models as electric fields to simulate neural activities using NEURON [47] within the Python environment. MRG models of various diameters represented different nerve fiber groups: Aα (16.0 μm), Aβ (10.0 μm), Aγ (7.3 μm), and Aδ (5.7 μm) [48]. Square pulses with a duration of 200 μs constituted the stimulation. A fiber was considered recruited if an action potential propagated its entire length. For each nerve root, ten trajectories were generated whose entries at the spinal cord were distributed evenly along the axial direction. Recruitment ratios of nerve roots and white matter were calculated for different stimulation amplitudes by scaling extracellular voltages under quasi-static assumptions. The activation threshold for a specific nerve root was defined as the stimulation amplitude at which 10% of Aα fibers were activated. Threshold maps were generated by traversing cathode positions and stimulation amplitudes (Fig. 2f).

To investigate the limits of spatial selectivity using EES, we varied the sizes of the target regions and conducted simulation analyses (Fig. 3f). Virtual circular target regions with diameters ranging from 0.5 mm to 2.5 mm and a thickness of approximately 2 mm were defined as targeted regions. The center of the contacts was positioned directly opposite the center of the circular target region. The multipolar configuration, which demonstrated superior performance compared to other configurations, was used for these simulations. We adjusted the spacing between contacts (ranging from 0.5 mm to 5 mm) and increased stimulation amplitudes to achieve optimal selectivity index for different target regions.

III. RESULTS

A. Preferred stimulation positions for different spinal cord segments

We conducted simulations of EES based on the developed human spinal cord FEM model and calculated selectivity indices using two calculation criteria for each stimulation configuration. Curves illustrating two selectivity indices with increasing stimulation amplitudes are shown in Fig. 2b and Fig. 2c. As stimulation amplitudes increase, the activated area in the first affected segment expands, leading to an increase in the selectivity index. However, beyond a certain stimulation amplitude threshold, gradual activation of other segments occurs, causing the selectivity index to decrease. By varying the position of stimulating contacts and stimulation amplitudes, selectivity index maps were plotted for L1-S2 segments, directly linked to lower limb motor function. Utilizing a monopolar configuration consistently achieved maximal selectivity index when contacts directly over the target segment were active (Fig. 2e). Larger segments (such as L1, L2) afford more flexibility in stimulation locations, while achieving a high selectivity index for smaller segments (e.g., S1, S2) requires a narrower range of effective locations. The AF-based selectivity index exhibits lower left-right selectivity compared to the Potential-based selectivity index, resulting in a lower average. An intriguing phenomenon occurs where the maximal AF-based SI for the left L1 segment is achieved when placing the cathode over the contralateral region. This anomaly is attributed to the generation of a region characterized by low activation function values, situated axially at a distance of 30 mm from the cathode, under monopolar stimulation. This diminishes the activation ratio of the target segment when placing the cathode over the left L1 segment. The Potential-based selectivity index demonstrates superior left-right selectivity and a more intuitive spatial distribution. Through the threshold maps, placing the contacts along the nerve root trajectory could activate the targeted nerve root with lower amplitudes (Fig. 2f). However, there are overlaps in the areas where exhibits low stimulation thresholds of different nerve roots, which indicates it is easy to activate non-targeted nerve roots with monopolar EES configurations.

B. Spatial selectivity of different stimulation configurations in EES

We computed the selectivity index of various stimulation configurations for EES, with a focus on the activation of different segments of the spinal cord. Most clinically used electrode arrays of EES feature staggered rows of contacts. We emulated this scenario and compared the spatial selectivity of several typical stimulation configurations. In addition to the monopolar configuration (ground the IPG case), anodes were added near the cathode to steer the current [13], [49] (Fig. 3a). Stimulation positions and amplitudes were adjusted to maximize the selectivity index in the epidural space. The multipolar configuration with 1 mm spacing between contacts demonstrates an advantage

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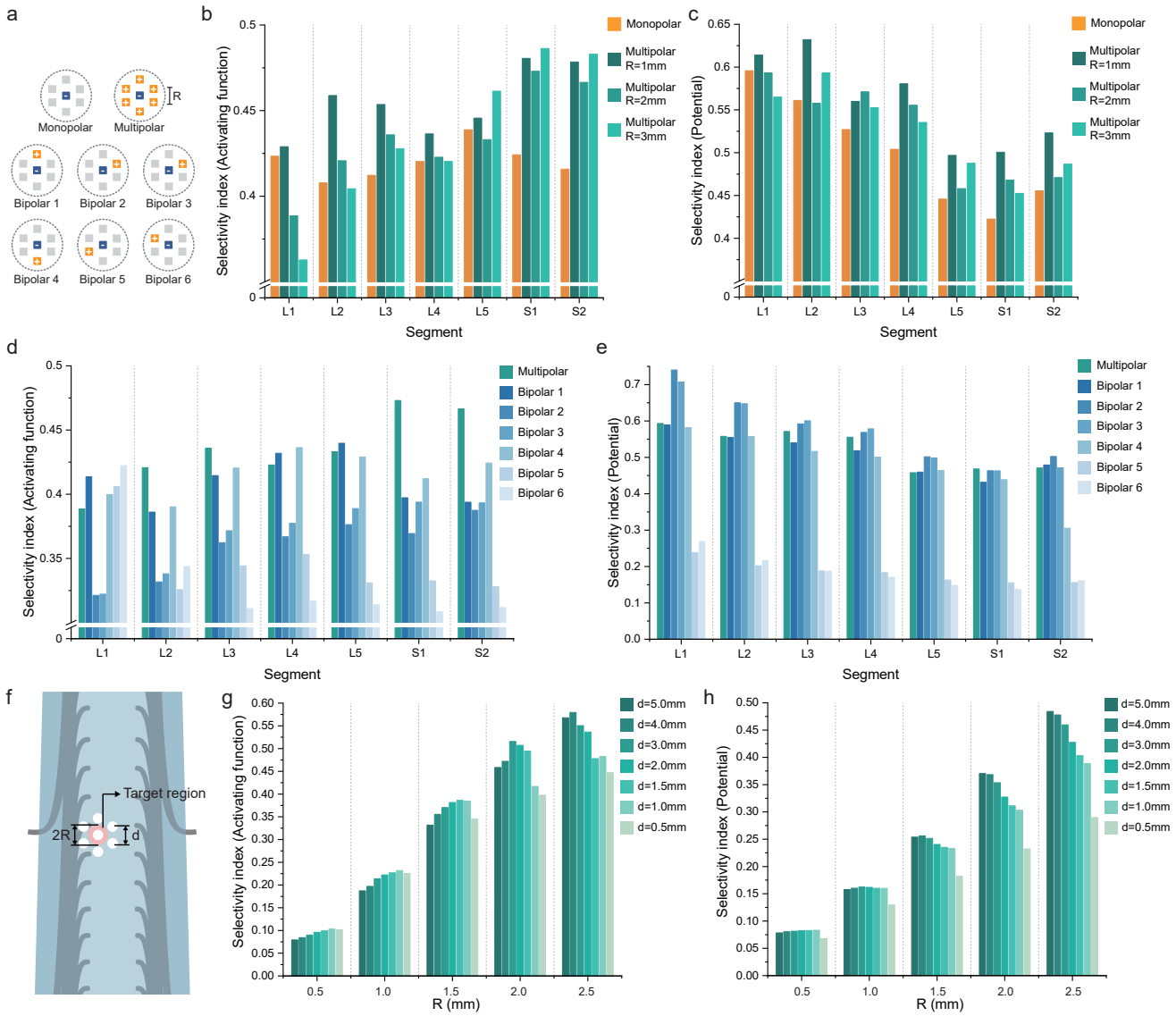


Fig. 3. Selectivity index of different configurations for EES. (a), Illustration of different stimulation configurations. (b), Selectivity index (based on activating function) of monopolar configuration and multipolar configurations. The spacing between contacts of multipolar configurations varies from 1 mm to 3 mm. (c), Selectivity index (based on potential) of monopolar configuration and multipolar configurations. (d), Selectivity index (based on activating function) of multipolar configurations and bipolar configurations. The spacing between contacts is 2 mm. (e), Selectivity index (based on potential) of multipolar configurations and bipolar configurations. (f), Illustration of multipolar EES for the target region. The radius of the target region is 2R and the thickness is around 2 mm. The spacing between contacts is d. (g), Selectivity index (based on activating function) of electrode arrays with different contact spacing when stimulating target regions with different radii. (h), Selectivity index (based on potential) of electrode arrays with different contact spacing when stimulating target regions with different radii.

over the monopolar configuration in both selectivity indices, indicating that adding anodes around the cathode can effectively limit the distribution of the electric field (Fig. 3b, c). However, if the contact spacing is too large, this advantage would be diminished.

The relative position of the anode and cathode influences the direction of the stimulation current [23]. We compared six bipolar configurations with different orientations. Simulation results indicate that axial orientations (Bipolar 1 and Bipolar 4 in Fig. 3a) achieve higher AF-based selectivity indices, while bipolar configurations with other orientations are more likely to perform worse than the multipolar configuration (Fig. 3d). However, a higher Potential-based selectivity index can be achieved by placing the cathode further away from the midline than the anode (Bipolar 2 and Bipolar 3 in Fig. 3a), whereas placing the cathode closer to the midline than the anode would decrease the calculated spatial selectivity (Fig. 3e). This occurs because an anode placed between the cathode and the midline reduces the flow of the current to the contra-lateral regions.

C. Limit in spatial selectivity of EES

We adjusted the size of the target region and calculated corresponding selectivity indices for various contact spacings

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(Fig. 3f). We observed that the maximum selectivity index was not consistently achieved with the smallest contact spacing. Larger target regions could be activated more selectively with larger contact spacing, as it can be challenging to fully activate a region with tightly-distributed contacts (Fig. 3g, h). Notably, a contact spacing of 1 mm outperformed 0.5 mm spacing across nearly all target regions, even for the smallest target region (with a radius of 0.5 mm), which indicates that decreasing contact spacing lower than 1 mm would not help improve the selectivity of stimulation.

IV. CONCLUSIONS

Epidural electrical stimulation has demonstrated significant potential in pain management and motor function rehabilitation. The electric fields induced by implanted electrode arrays are strongly correlated with the stimulation effects, making it crucial to quantitatively understand the spatial characteristics of induced electric fields for different configurations. In this study, we developed an average model of the human spinal cord and simulated EES using finite element methods and neural activity simulations. Through automated modeling and simulation, we systematically investigated neural activation and spatial selectivity across multiple levels of the spinal cord. Our results quantitatively confirmed the superiority of multipolar stimulation over monopolar stimulation in terms of spatial selectivity and amplitude regulation. Additionally, we compared stimulation efficacy of electrode arrays with varying contact spacings and topologies, offering insights for the impact of parameter configurations of EES. Moreover, we explored the potential limits of spatial selectivity when using electrode arrays positioned outside the dura mater. Our results indicate simply increasing contact density would not efficiently improve the precision and selectivity of EES due to the existence of highly conductive CSF. This study provides a valuable quantitative reference for optimizing interface design and therapeutic settings, with implications for enhancing the precision and efficacy of EES in clinical applications.

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